# NMTC Problem-Set 2

**Instructions:** For this problem-set, it is assumed that you know the following topics to some degree.

1. Angle Chasing
2. Basic Combinatorics
3. Basic Divisibility
4. Proof by Contradiction
5. Basic methods of solving Functional Equations, including induction, injectivity/surjectivity, etc.

This problem-set is meant to be of the same level as the NMTC descriptive questions and above that. Problems marked at the end with (*) are a bit more tough (and fun!), while problems marked with (**) are tougher, and so on.
These problems are meant to build your problem-solving skills, and as such, they are open-book. You are encouraged to use any textbook, video, or other online or offline resource as long as it doesn't give away the complete solution.
The sources to each applicable problem will be revealed with the solutions.

# Problem 1: Angle Chasing...again

Note: If you want a real challenge, attempt part ($d$) while skipping the other parts without reading them.

Let $ABC$ be a triangle with circumcircle $\Gamma$. Let $D \neq A$ be the second intersection of the internal bisector of $\angle BAC$ with $\Gamma$. We define $E$ to be the intersection of line $CD$ with the line perpendicular to $BC$ through $B$, and $\omega$ to be the circumcircle of $ADE$. The line parallel to $AD$ passing through $E$ intersects $\omega$ at $F \neq E$. Moreover, let the tangents to $\Gamma$ at $A$ and $C$ intersect at $T$.

(a) Prove that $DE = DC$.

(b) Show that $FADE$ is an isosceles trapezoid. An isosceles trapezoid consists of two opposite sides parallel and the other two opposite side equal in length.

(c) Prove that $\triangle FAT \cong \triangle DCT$

(d) Show that $FT$ is tangent to $\omega$.*

# Problem 2: Counting

1. A club has 11 men and 12 women. A committee is chosen so that the number of women on the committee is one more than the number of men. (Committees can have as few as 1 or as many as 23 members.) Let $N$ be the number of such committees that can be formed. Find the sum of the prime factors of $N$.

2. Compute the number of sets of three distinct elements that can be chosen from the set $S = \{2^1, 2^2, 2^3, ..., 2^{2000}\}$, such that the three elements form a geometric progression. For example, the set $\{2^1, 2^2, 2^3\}$ has its elements in a geometric progression with common ratio 2.

   For an added challenge, can you find a general formula for the number of such sets if the last term in the set S is replaced by $2^k$, where $k$ is any natural number? What if the base 2 is replaced by any natural number $n$? What if the base is $n$ and the last term is $n^k$? (*)

# Problem 3: Some Divisibility

Let $a$, $b$ be integers with $1 < b < a$. Suppose that the quantity $(a^2 - 1)(b^2 - 1)$ is a perfect square.

(a) Prove that $a + b$ is composite.

(b) Prove that if $a - b$ is prime, then $6ab - 3$ is a perfect square.

# Problem 4: Some Algebra!

1. Let $m$, $n$ be positive integers. An $m \times n$ table has a real number written in all of its cells. We call such a table a *product* table if each cell's number equals the product of all the numbers in its neighboring cells. A cell neighbors another if it shares an edge.

   (a) Find all $2 \times 2$ *product* tables.

   (b) Find all $2 \times 3$ *product* tables.

   (c) Find all $3 \times 3$ *product* tables.

2. Find all functions $f : \mathbf{R} \mapsto \mathbf{R}$ satisfying:

$$f(f(x) + y) = 2x + f(f(y) - x). \qquad (**)$$